

Governance / Constitution Debt Status Audit

Date: 2026-06-14 **Scope:** Read-only verification of every §6.*-debt entry + CM-* paper-only/owed gate in root CLAUDE.md, cross-checked against the actual code/scripts/hooks at HEAD 357f867a. **Method:** Each row's "actual-status" is verified against a real artifact (file existence, hook line, verify-all wiring), not against the CLAUDE.md prose. Where CLAUDE.md prose disagrees with the artifact, the row is flagged STALE. **Anti-bluff (§11.4.6):** Every claim is cited to a file/line or "absent". No guessed status.

Debt-item status table

debt-id	requires	actual-status	evidence (file:line / absent)
§6.X-debt	Emulator PID inside podman/docker container managed by pkg/runtime; --runner=containerized for gate runs on every OS	PARTIAL	submodules/containers/pkgcontainerized.go exists with kvmDevicePath/kvmAvailabl (containerized.go:14-35); path on macOS still resolves RESOLVED_RUNNER="host-dir (scripts/run-challenge-matrix.sh:198,215). Linux x path works; macOS gate-host B Superseded in practice by §6.A
§6.Y-debt	Pre-push hook rejects code-touching commit after a distribute that did not bump versionCode	CLOSED	.githooks/pre-push:128-15 implements §6.Y bump-first or last-version-debug. CLAU lists it OWED → STALE.
§6.Z-debt	firebase-distribute Phase-1 Gate 6 (evidence file: matching SHA + ≤24h + BUILD SUCCESSFUL) + pre-push check + tests/ firebase/ hermetic test	PARTIAL	Pre-push Check 7 enforces evidence presence on pointer advance (. pre-push:159-193). But scri firebase-distribute.sh has Gate 6 that validates SHA/time BUILD-SUCCESSFUL — only reference (firebase-distribute.sh:40-41). test exists but tests version-monoton not the §6.Z evidence gate (tes firebase/).
§6.AA-debt	Default --debug-only; reject --release-only without companion	CLOSED	firebase-distribute.sh:46 MODE="debug" default; :199-2 release-only when

debt-id	requires	actual-status	evidence (file:line / absent)
	debug evidence; split last-version- {debug, release}; tests/firebase/ hermetic test		last-version-debug < curr pointers :150,171. CLAUDE.m PARTIAL → STALE.
§6.AB-debt	Detekt/go-vet rule for per-feature anti-bluff completeness (rendering/ state-machine/gating)	OPEN	No Detekt rule; config/detek only detekt.yml (config/det The related scripts/check-cl discrimination.sh exists+ST covers Challenge falsifiability r the §6.AB completeness checkl
§6.AC-debt	Detekt rule config/ detekt/lava-non- fatal-required.kts + go-vet lava-api-go/ scripts/check-non- fatal-coverage.sh + pre-push + tests/non- fatal-coverage/	OPEN	config/detekt/lava-non-fa required.kts ABSENT; lava scripts/check-non-fatal-c ABSENT; tests/non-fatal- ABSENT. A bash scanner scri non-fatal-coverage.sh exist only, STRICT) but the Go vet + custom rule + hermetic tests are
§6.AD-debt (1) per- submodule pointer propagation	## INHERITED FROM block in all per-scope CLAUDE.md/ AGENTS.md/ CONSTITUTION.md	CLOSED	scripts/check-constitutio 6.AD(4) enforces 54 in-scope f helix-constitution-gates.
§6.AD-debt (2) CM- COMMIT-DOCS- EXISTS	verify referenced docs in commit messages resolve	CLOSED	scripts/check-commit-docs exists + wired in verify-all (scr verify-all-constitution- rules.sh:190). CLAUDE.md → STALE.
§6.AD-debt (2) CM- SCRIPT-DOCS-SYNC	script [?] docs/scripts/ *.md sync gate	CLOSED	scripts/check-script-docs exists + verify-all :187; AND p Check 9 (.github/hooks/pre-pus CLAUDE.md lists OWED → S
§6.AD-debt (2) CM- SUBAGENT- DELEGATION-AUDIT	subagent delegation audit trail	CLOSED (scanner)	scripts/check-subagent-de audit.sh exists + verify-all :1 + CLAUDE.md call it paper-on
§6.AD-debt (2) CM- COMMIT-DOCS- EXISTS/others equivalence- mapped	CM-ITEM-STATUS/TYPE- TRACKING, CM-FIXED- COLUMN-ALIGNMENT, operator-blocked gates	CLOSED-BY- EQUIVALENCE	§6.AD.3 Path B maps these to evidence/ ledgers (docs/hel constitution- gates.md:14,16,19-22,36-3 Intentional; no scanner owed.
§6.AD-debt (5) build-resource stats tracker	CM-BUILD-RESOURCE- STATS-TRACKER	CLOSED	scripts/build-stats-samp scripts/build-stats-repor registry exist (docs/helix-co gates.md:18). CLAUDE.md § still lists OWED → STALE.
§6.AE-debt	check-challenge- coverage.sh strict-	CLOSED (strict)	scripts/check-challenge- coverage.sh:39 STRICT="\$ {LAVA_CHALLENGE_COVERAGE_

debt-id	requires	actual-status	evidence (file:line / absent)
	mode flip + per-feature backfill		(STRICT default); verify-all inv (:163). Gate-doc says "advisor Runner host-gap is §6.X/§6.AH
§6.AF-debt (2) chaos/stress §11.4.85 (LVA-7)	runnable chaos/stress beyond phase-1 lava-api-go	PARTIAL	scripts/run-chaos-stress.s lines, scaffold); docs/chaos-s DESIGN.md + EVIDENCE-phase NOT wired into verify-all (no c scripts/verify-all-consti rules.sh). Phase-2+ owed.
§6.AF-debt (LVA-6) codegraph own-org submodule inclusion	§11.4.79 — index own-org submodule SOURCE	CLOSED	.codegraph/config.json:3 records LVA-6 CLOSED 2026-submodule files indexed; cross-probe resolves); blanket submo exclude removed. CLAUDE.m OWED → STALE.
§6.AF-debt (4) per-clause CM-COVENANT-114-*PROPAGATION gates	literal-anchor scan gates for §11.4.79–106	OPEN	Only CM-COVENANT-114-16-P row present (docs/helix-con gates.md:14); no per-clause p scanner for the new 79–106 and
§6.AF-debt (5) §11.4.80 codegraph sync automation	auto git submodule init before index	OPEN	Precondition documented in .c config.json:3 but no automa runs git submodule init bef codegraph index.
§6.AG (no separate -debt)	Containers-driven emulators, no live ADB device	PARTIAL	Policy is in CLAUDE.md §6.A enforced via run-challenge-r but the no-host-direct requirem superseded/tightened by §6.AH by §6.AH-debt.
§6.AH-debt	Container/VM emulator path that BOOTS on macOS (no-KVM TCG mode + macOS /proc- reap fix in Containers submodule); no host- direct fallback	OPEN	containerized.go hard-requi for acceleration (containerized.go:19–21, 3 no-KVM software-emulation m challenge-matrix.sh:215 st macOS→host-direct. Incide .lava-ci-evidence/sixth-l incidents/2026-06-03-emul offline.json (present). §6.Z + client gate honestly BLOCKE macOS host.
§6.AI-debt (1) §11.4.128 device recorder	background Genymotion/emulator recorder w/ deterministic layout	PARTIAL/ CLOSED?	scripts/record-device-ses exists; tests/device-record Whether it implements the full YYYY-MM-DD/<state-hash>/ gitignore/codegraph-exclude w verified — marked PARTIAL p check.
§6.AI-debt (2) §11.4.140 LAYER 2 action-prefix hook	UserPromptSubmit hook + generated slash commands	CLOSED (hook)	scripts/hooks/action-pref expand.sh exists + wired in .c settings.json:14–22

debt-id	requires	actual-status	evidence (file:line / absent)
			(UserPromptSubmit); consti actions/registry.yaml pres Generated per-agent slash comm verified. CLAUDE.md lists OW STALE for the hook portion.
§6.AI-debt (3) §11.4.141 thin- index restructure + token harness	de-dup 90K-token CLAUDE.md to index lines	OPEN	No evidence of a thin-index res CLAUDE.md remains a full-body
§6.AI-debt (4) per- clause CM- COVENANT-114- {128..141}- PROPAGATION gates	literal-anchor gates for §11.4.128-141	OPEN	Not present in docs/helix-co gates.md gate inventory.
CM-COVERAGE- LEDGER	§11.4.25 — strict-flip after baseline (Phase 7- debt)	CLOSED (strict)	scripts/check-coverage-le STRICT=\$ {LAVA_COVERAGE_LEDGER_STP verify-all :156 invokes --stri + CLAUDE.md mark it "advise flip OWED" → STALE.
CM-LVA-TICKETS- SYNC → CM- WORKABLE-ITEMS- SYNC	canonical workable- items binary; strict-flip	PARTIAL	scripts/check-workable-it + verify-all :202. Gate-doc stil retired CM-LVA-TICKETS-SYNC check-lva-tickets.sh (docs constitution-gates.md:25) CLAUDE.md §6.AF notes mig canonical done. Upstream upda block + export pipeline OWED blocking).
CM- COVENANT-114-16- PROPAGATION + item-tracking paper- only gates	§11.4.15/16/19/21 trackers	CLOSED-BY- EQUIVALENCE	Intentional Path-B mapping (do constitution-gates.md:14, true debt.

STALE-claim findings (CLAUDE.md/gate-doc says OPEN but artifact is CLOSED)

These are documentation drift, not real work. A master plan should NOT schedule work for them — only doc corrections:

1. **§6.Y-debt** — pre-push Check 6 is live (.github/hooks/pre-push:128). CLAUDE.md lists OWED.
2. **§6.AA-debt** — default flipped to --debug-only; release-only companion check live (firebase-distribute.sh:46,199). CLAUDE.md lists PARTIAL.

3. **§6.AD-debt items 2** — `check-commit-docs-exists.sh`, `check-script-docs-sync.sh`, `check-subagent-delegation-audit.sh` all exist + verify-all-wired + pre-push Check 9. Gate-doc marks several ⚠ paper-only.
4. **§6.AD-debt item 5** — `build-stats` tracker shipped (`build-stats-sample.sh`/`build-stats-report.sh`). CLAUDE.md still lists OWED.
5. **§6.AE-debt** — `check-challenge-coverage.sh` is STRICT-default (:39). Gate-doc says advisory.
6. **CM-COVERAGE-LEDGER** — STRICT-default (`check-coverage-ledger.sh:41`, `verify-all --strict`). CLAUDE.md + gate-doc say advisory / Phase-7-debt OWED.
7. **LVA-6 (codegraph submodules)** — CLOSED per `.codegraph/config.json:3`. CLAUDE.md §6.AF lists OWED.
8. **§6.AI LAYER 2 action hook** — wired in `.claude/settings.json:14`. CLAUDE.md lists OWED.
9. **docs/helix-constitution-gates.md itself is stale** — "Last updated 2026-05-16"; predates §6.AF/AG/AH/AI (May 31 – Jun 9). Several rows describe pre-strict-flip state.

Truly-open debts (real work to schedule)

debt-id	one-line	effort	blocking
§6.AH-debt	macOS container/VM (TCG no-KVM) emulator path that actually boots; no host-direct fallback	L	YES
§6.X-debt	container-bound emulator gate path on macOS (folds into §6.AH)	L	YES
§6.AC-debt	Detekt <code>lava-non-fatal-required.kts</code> + Go-vet <code>check-non-fatal-coverage.sh</code> + <code>tests/non-fatal-coverage/</code>	M	no
§6.AB-debt	Detekt/go-vet per-feature anti-bluff completeness rule	M	no
§6.Z-debt (residual)	runtime Gate 6 inside <code>firebase-distribute.sh</code> (SHA/24h/BUILD-SUCCESSFUL) + hermetic test	M	partial
§6.AF-debt chaos (LVA-7)	chaos/stress §11.4.85 beyond phase-1; wire into <code>verify-all</code>	M	no
§6.AI-debt §11.4.141	thin-index restructure of CLAUDE.md + token-reduction harness	L	no
§6.AF/AI per-clause CM-COVENANT-* PROPAGATION gates	literal-anchor scanners for §11.4.79–141	S	no
§6.AF-debt §11.4.80	auto git submodule init before codegraph index	S	no
§6.AI-debt §11.4.128	verify/complete deterministic device-recorder layout	S	no

Prioritized recommendations

1. **§6.AH-debt + §6.X-debt (TOP — both BLOCKING)**. Single coordinated effort in the Containers submodule: add a no-KVM TCG software-emulation runner + the macOS /proc-reap fix so the container/VM emulator path boots on darwin/arm64. Until this lands, EVERY §6.AE matrix gate, §6.Z release canary, and tag-time attestation on the operator's macOS host is honestly BLOCKED — this is the single highest-leverage debt because it unblocks the entire release pipeline.
2. **§6.Z-debt residual (firebase Gate 6)**. Add the runtime evidence gate inside `firebase-distribute.sh` (the pre-push Check 7 only fires on the commit that advances the pointer, not at distribute time). Cheaper than #1 and closes a real `distribute-faulty-version` vector.
3. **§6.AC-debt (non-fatal telemetry lint)**. Build the Detekt custom rule + Go-vet check + `tests/non-fatal-coverage/`. Currently only a Kotlin bash scanner exists; the Go side has zero enforcement. Pairs naturally with §6.AB-debt (same Detekt-rule-authoring effort).

Doc-hygiene action (not scheduled work, but should precede the plan): correct the 9 STALE claims above in `CLAUDE.md` + regenerate `docs/helix-constitution-gates.md` (stale since 2026-05-16) so the master plan is built on accurate status. Several debts are already done.

Open vs closed count: Of ~27 distinct debt/gate items examined: **10 truly-open** (2 BLOCKING, 1 partial-blocking, 7 non-blocking), **3 PARTIAL**, ****~14 CLOSED** or closed-by-equivalence** (9 of which are STALE-marked-open in `CLAUDE.md/gate-doc`).